

## Initiating Search

November 10, 2025, 10:29 PM

 Search:

CAS Lexicon Search:

C-H bond activation - Preferred Concept

## Search Tasks

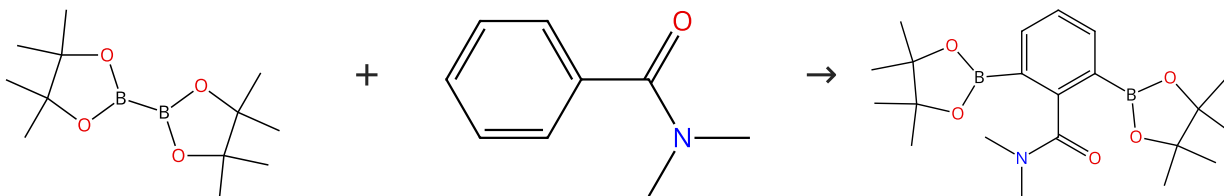
| Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Result Type                                                                                   | View                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------|
| Returned Reference Results from CAS Lexicon (13,133)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  References | <a href="#">View Results</a> |
| Exported: Retrieved Related Reaction Results + Filters (1,085)<br><br>Filtered By:<br><br>Yield: 90-100%, 80-89%, 70-79%<br>Catalyst: Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, Di-μ-chlorobis[(1,2,5,6-η)-1,5-cyclooctadiene]diiridium, Di-μ-chlorotetrakis[(1,2-η)-cyclooctene]diiridium, Chloro[(1,2,5,6-η)-1,5-cyclooctadiene](1,10-phenanthroline-κ <sup>N1</sup> ,κ <sup>N10</sup> )iridium, [(1,2,5,6-η)-1,5-Cyclooctadiene][(1,2,3,3a,7a-η)-1 <i>H</i> -inden-1-yl]iridium, Zirconium, bis[μ <sub>3</sub> -[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κ <sup>N1</sup> ,κ <sup>N10</sup> ]benzoato(2-)-κ <sup>O</sup> :κ <sup>O'</sup> ]]bis[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetrakis[μ-[2"-nitro[1,1':4',1"':4",1"':quaterphenyl]-4,4"-dicarboxylato(2-)-κ <sup>O4''</sup> :κ <sup>O4'''</sup> ]]tetra-μ <sub>3</sub> -oxohexa-, Zirconium, tetrakis[μ-[(2 <i>E</i> )-3-[4'-[(1 <i>E</i> )-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κ <sup>O</sup> :κ <sup>O'</sup> ]]bis[μ <sub>3</sub> -[(2 <i>E</i> )-3-[5'-[(1 <i>E</i> )-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κ <sup>N1</sup> ,κ <sup>N1'</sup> ]-2-propenoato(2-)-κ <sup>O</sup> :κ <sup>O'</sup> ]]bis[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetra-μ <sub>3</sub> -oxohexa-, Zirconium, hexakis[μ <sub>3</sub> -[(2 <i>E</i> )-3-[5'-[(1 <i>E</i> )-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κ <sup>N1</sup> ,κ <sup>N1'</sup> ]-2-propenoato(2-)-κ <sup>O</sup> :κ <sup>O'</sup> ]]hexakis[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetra-μ <sub>3</sub> -oxohexa-, Stereoisomer of [(1,2-η)-1-cyclooctene]hydro[8-(hydroxy-κ <sup>O</sup> )-2-quinolinecarboxylato(2-)-κ <sup>N1</sup> ,κ <sup>O2</sup> ](pyridine)iridium |  Reactions  | <a href="#">View Results</a> |

# Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (125)

 Suppliers (68)

31-614-CAS-37278699

Steps: 1 Yield: 100%

**Iridium-Catalyzed ortho-Selective Borylation of Aromatic Amides Enabled by 5-Trifluoromethylated Bipyridine Ligands**

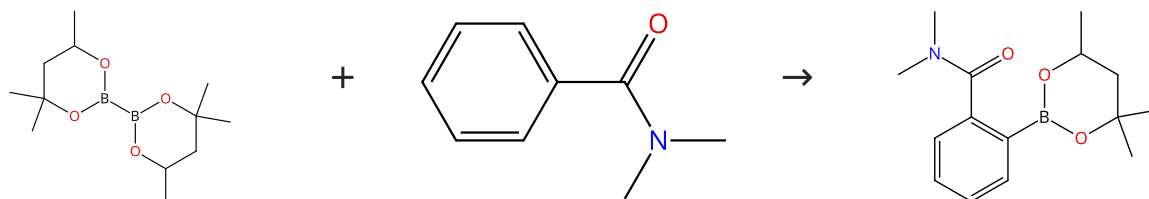
By: Marcos-Atanes, Daniel; et al

Angewandte Chemie, International Edition (2023), 62(18), e202214510.

1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 5,5'-Bis(trifluoromethyl)-2,2'-bipyridine  
**Solvents:** Tetrahydrofuran; 1.5 h, 65 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (75)

 Suppliers (68)

31-614-CAS-37278654

Steps: 1 Yield: 100%

**Iridium-Catalyzed ortho-Selective Borylation of Aromatic Amides Enabled by 5-Trifluoromethylated Bipyridine Ligands**

By: Marcos-Atanes, Daniel; et al

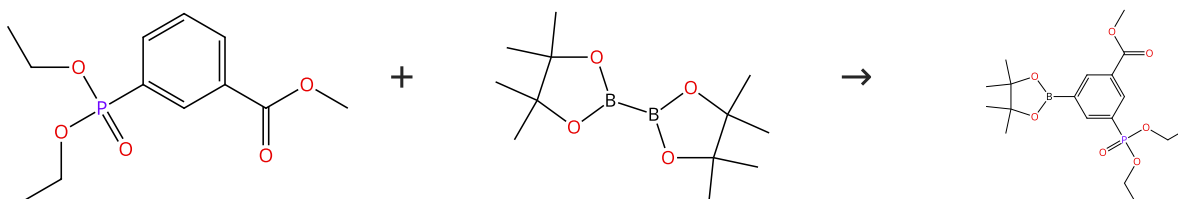
Angewandte Chemie, International Edition (2023), 62(18), e202214510.

1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 5-(Trifluoromethyl)-2,2'-bipyridine  
**Solvents:** Tetrahydrofuran; 24 h, 65 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

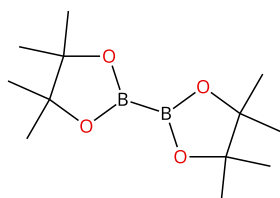

 Suppliers (4)

 Suppliers (125)

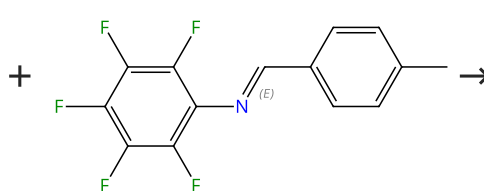
|                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31358923                                                                                                                                             | Steps: 1 Yield: 100% | <b>The Synthesis of Biarylmonophosphonates via Palladium-Catalyzed Phosphonation, Iridium-Catalyzed C-H Borylation, Palladium-Catalyzed Suzuki-Miyaura Cross-Coupling</b><br><br>By: Doherty, Simon; et al<br><br>Catalysis Letters (2022), 152(2), 398-413. |
| 1.1 <b>Catalysts:</b> 3,4,7,8-Tetramethyl-1,10-phenanthroline, Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium<br><b>Solvents:</b> Hexane; 18 h, 70 °C |                      |                                                                                                                                                                                                                                                              |
| Experimental Protocols                                                                                                                                          |                      |                                                                                                                                                                                                                                                              |

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%

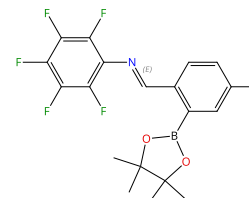


Suppliers (125)



Double bond geometry shown

Supplier (1)

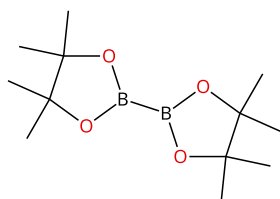


Double bond geometry shown

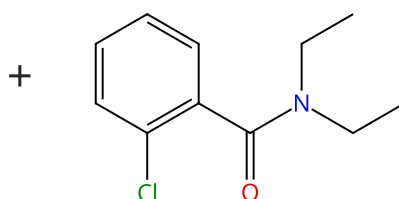
|                                                                                                                                                             |                      |                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-142-CAS-15450032                                                                                                                                         | Steps: 1 Yield: 100% | <b>Iridium-catalyzed ortho-C-H borylation of aromatic aldimines derived from pentafluoroaniline with bis(pinacolate)diboron</b><br><br>By: Sasaki, Ikuo; et al<br><br>Organic & Biomolecular Chemistry (2014), 12(13), 2041-2044. |
| 1.1 <b>Catalysts:</b> Tris(pentafluorophenyl)phosphine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium<br><b>Solvents:</b> Mesitylene; 10 min, rt |                      |                                                                                                                                                                                                                                   |
| 1.2 <b>Reagents:</b> Norbornene; 7 h, 120 °C                                                                                                                |                      |                                                                                                                                                                                                                                   |
| Experimental Protocols                                                                                                                                      |                      |                                                                                                                                                                                                                                   |

Scheme 5 (1 Reaction)

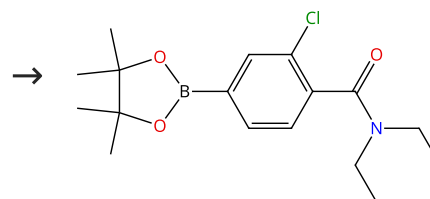
Steps: 1 Yield: 100%



Suppliers (125)



Suppliers (39)

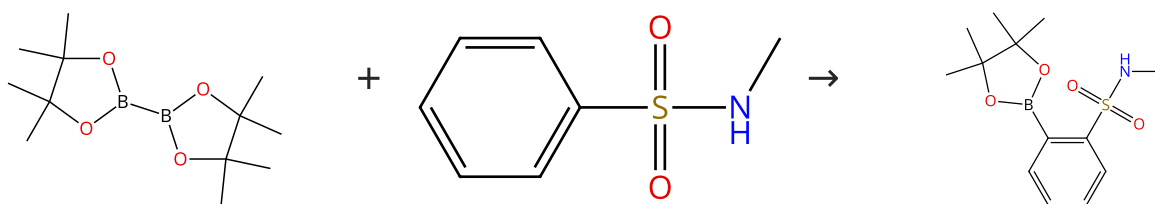


Suppliers (7)

|                                                                                                                                                                                                             |                      |                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-142-CAS-17033452                                                                                                                                                                                         | Steps: 1 Yield: 100% | <b>para-Selective C-H Borylation of (Hetero)Arenes by Cooperative Iridium/Aluminum Catalysis</b><br><br>By: Yang, Lichen; et al<br><br>Angewandte Chemie, International Edition (2017), 56(17), 4853-4857. |
| 1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,2'-bipyridine<br><b>Solvents:</b> Hexane; 30 min, 60 °C; 60 °C → rt |                      |                                                                                                                                                                                                            |
| 1.2 <b>Catalysts:</b> Bis[2,6-bis(1,1-dimethylethyl)-4-methylphenolato]methylaluminum<br><b>Solvents:</b> Hexane; 18 h, rt                                                                                  |                      |                                                                                                                                                                                                            |
| 1.3 <b>Reagents:</b> Ethyl acetate; rt                                                                                                                                                                      |                      |                                                                                                                                                                                                            |
| Experimental Protocols                                                                                                                                                                                      |                      |                                                                                                                                                                                                            |

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (60)

Suppliers (29)

31-614-CAS-37278652

Steps: 1 Yield: 100%

**Iridium-Catalyzed ortho-Selective Borylation of Aromatic Amides Enabled by 5-Trifluoromethylated Bipyridine Ligands**

By: Marcos-Atanes, Daniel; et al

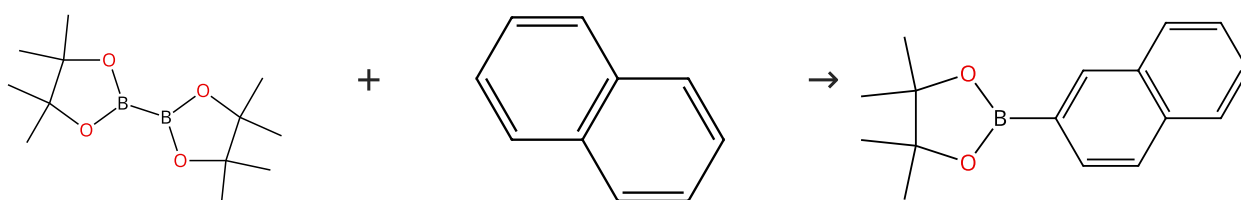
Angewandte Chemie, International Edition (2023), 62(18), e202214510.

1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 5-(Trifluoromethyl)-2,2'-bipyridine  
**Solvents:** Tetrahydrofuran; 24 h, 65 °C

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (134)

Suppliers (77)

31-142-CAS-17492167

Steps: 1 Yield: 100%

**C-H activation-based transformation of naphthalenes to 3-iodo-2-naphthylboronic acid derivatives for use in iterative coupling synthesis of helical oligo(naphthalene-2,3-diyl)s**

By: Yamamoto, Takeshi; et al

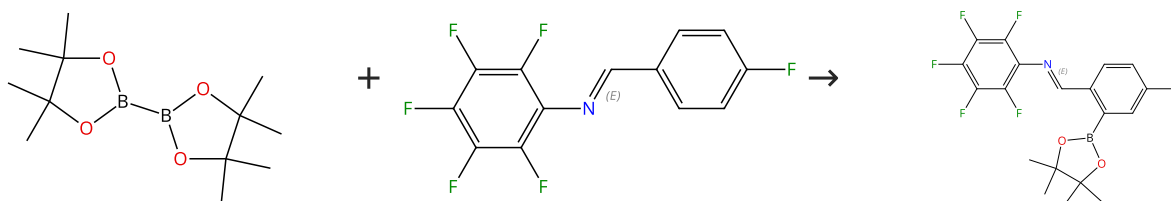
Bulletin of the Chemical Society of Japan (2017), 90(5), 604-606.

1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine  
**Solvents:** Cyclohexane; 18 h, 80 °C

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Double bond geometry shown

Double bond geometry shown

Supplier (1)

31-142-CAS-6825289

Steps: 1 Yield: 100%

**Iridium-catalyzed ortho-C-H borylation of aromatic aldimines derived from pentafluoroaniline with bis(pinacolato)diboron**

By: Sasaki, Ikuo; et al

Organic &amp; Biomolecular Chemistry (2014), 12(13), 2041-2044.

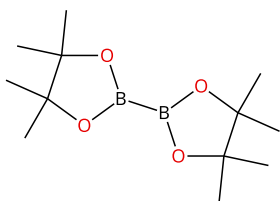
1.1 **Catalysts:** Tris(pentafluorophenyl)phosphine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium  
**Solvents:** Mesitylene; 10 min, rt

1.2 **Reagents:** Norbornene; 4 h, 120 °C

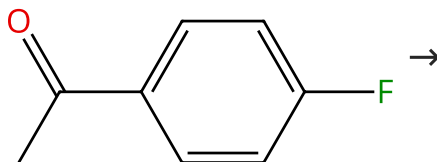
Experimental Protocols

Scheme 9 (1 Reaction)

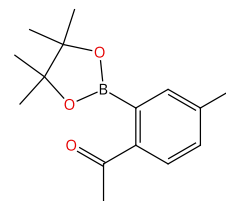
Steps: 1 Yield: 100%



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Suppliers (125)

Suppliers (93)

Suppliers (13)

31-142-CAS-17566936

Steps: 1 Yield: 100%

Achieving Site Selectivity in Metal-Catalyzed Electron-Rich Carbene Transfer Reactions from N-Tosylhydrazones

1.1 **Catalysts:** Triphenylarsine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Octane; 16 h, 120 °C

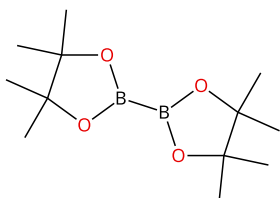
By: Su, Naijing; et al

Organic Letters (2017), 19(15), 3990-3993.

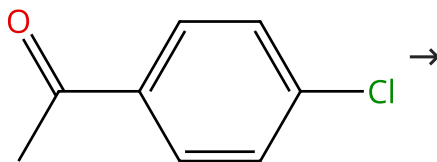
Experimental Protocols

Scheme 10 (1 Reaction)

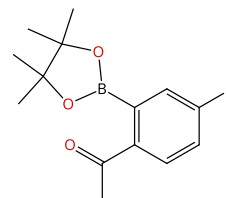
Steps: 1 Yield: 100%



+



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Suppliers (125)

Suppliers (67)

Suppliers (14)

31-142-CAS-17566935

Steps: 1 Yield: 100%

Achieving Site Selectivity in Metal-Catalyzed Electron-Rich Carbene Transfer Reactions from N-Tosylhydrazones

1.1 **Catalysts:** Triphenylarsine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Octane; 16 h, 120 °C

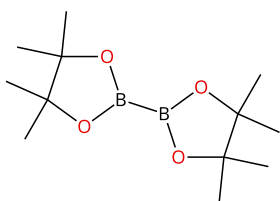
By: Su, Naijing; et al

Organic Letters (2017), 19(15), 3990-3993.

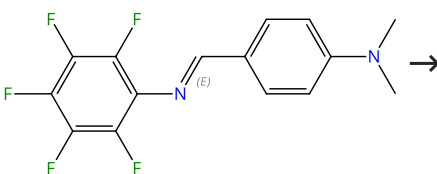
Experimental Protocols

Scheme 11 (1 Reaction)

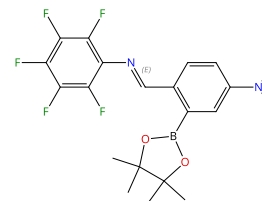
Steps: 1 Yield: 100%



+



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Suppliers (125)

Double bond geometry shown

Double bond geometry shown

31-142-CAS-12993233

Steps: 1 Yield: 100%

Iridium-catalyzed ortho-C-H borylation of aromatic aldimines derived from pentafluoroaniline with bis(pinacolato)diboron

1.1 **Catalysts:** Tris(pentafluorophenyl)phosphine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Mesitylene; 10 min, rt  
 1.2 **Reagents:** Norbornene; 8 h, 120 °C

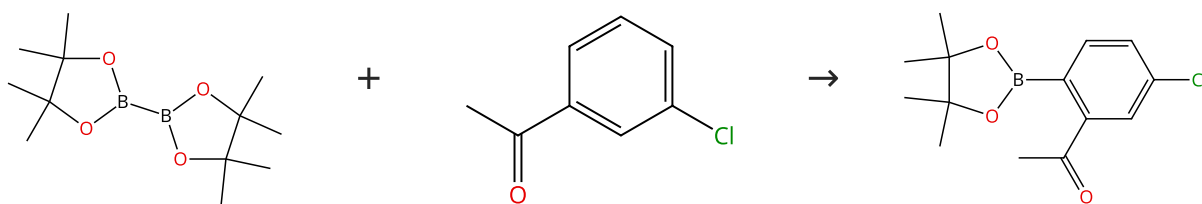
By: Sasaki, Ikuo; et al

Organic &amp; Biomolecular Chemistry (2014), 12(13), 2041-2044.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (58)

Suppliers (8)

31-142-CAS-17566940

Steps: 1 Yield: 100%

Achieving Site Selectivity in Metal-Catalyzed Electron-Rich Carbene Transfer Reactions from N-Tosylhydrazones

1.1 **Catalysts:** Triphenylarsine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Octane; 16 h, 120 °C

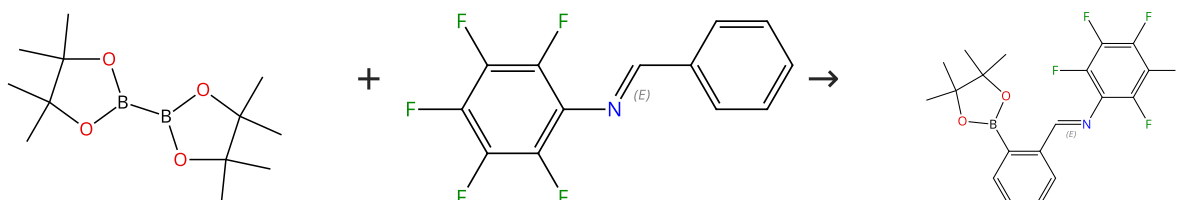
By: Su, Naijing; et al

Organic Letters (2017), 19(15), 3990-3993.

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Double bond geometry shown

Double bond geometry shown

Suppliers (2)

31-142-CAS-8775956

Steps: 1 Yield: 100%

Iridium-catalyzed ortho-C-H borylation of aromatic aldimines derived from pentafluoroaniline with bis(pinacolate)diboron

1.1 **Catalysts:** Tris(pentafluorophenyl)phosphine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Mesitylene, Water; 10 min, rt

By: Sasaki, Ikuo; et al

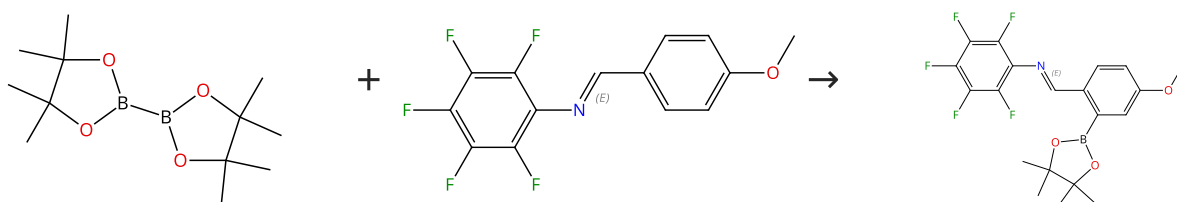
Organic &amp; Biomolecular Chemistry (2014), 12(13), 2041-2044.

1.2 **Reagents:** Norbornene; 4 h, 120 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Double bond geometry shown

Double bond geometry shown

Supplier (1)

31-142-CAS-4469479

Steps: 1 Yield: 100%

Iridium-catalyzed ortho-C-H borylation of aromatic aldimines derived from pentafluoroaniline with bis(pinacolate)diboron

1.1 **Catalysts:** Tris(pentafluorophenyl)phosphine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium  
**Solvents:** Mesitylene; 10 min, rt

By: Sasaki, Ikuo; et al

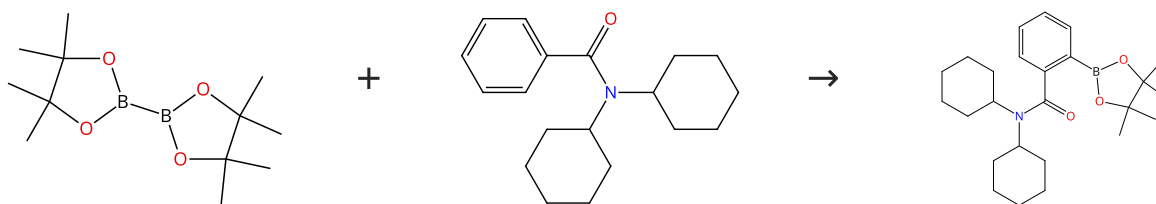
Organic &amp; Biomolecular Chemistry (2014), 12(13), 2041-2044.

1.2 **Reagents:** Norbornene; 3 h, 120 °C

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (11)

Suppliers (3)

31-614-CAS-37278635

Steps: 1 Yield: 100%

**Iridium-Catalyzed ortho-Selective Borylation of Aromatic Amides Enabled by 5-Trifluoromethylated Bipyridine Ligands**

By: Marcos-Atanes, Daniel; et al

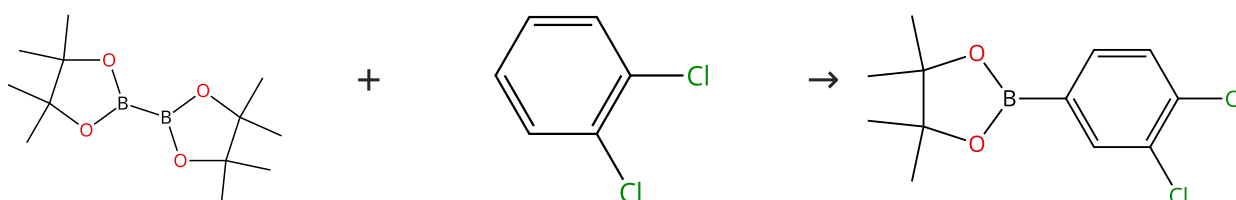
Angewandte Chemie, International Edition (2023), 62(18), e202214510.

1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 5-(Trifluoromethyl)-2,2'-bipyridine  
**Solvents:** Tetrahydrofuran; 24 h, 65 °C

Experimental Protocols

Scheme 16 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (132)

Suppliers (72)

31-142-CAS-228057

Steps: 1 Yield: 100%

**Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation**

By: Manna, Kuntal; et al

Journal of the American Chemical Society (2015), 137(7), 2665-2673.

1.1 **Catalysts:** Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthroline-3-yl-κN<sup>1</sup>,κN<sup>10</sup>]benzoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetrakis[μ-[2''-nitro[1,1':4',1''':4'',1''':4''',1''':4''']-quaterphenyl]-4,4''':dicarboxylato(2-)-κO<sup>4''</sup>:κO<sup>4'''</sup>]]]tetra-μ<sub>3</sub>-oxohexa-; 30 h, 115 °C

Experimental Protocols

31-142-CAS-13511089

Steps: 1 Yield: 100%

**Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation**

By: Manna, Kuntal; et al

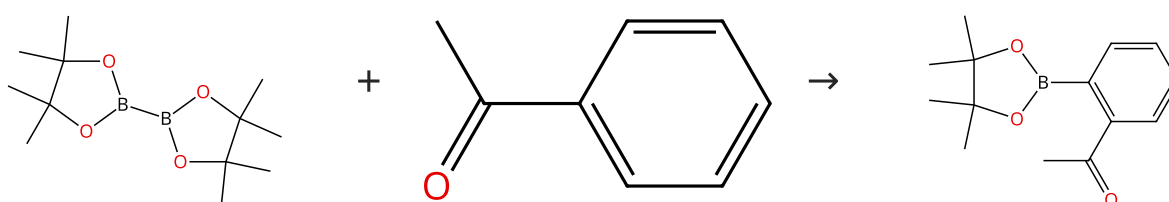
Journal of the American Chemical Society (2015), 137(7), 2665-2673.

1.1 **Catalysts:** Zirconium, tetrakis[μ-[(2E)-3-[4'-[(1E)-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κO:κO']]bis[μ<sub>3</sub>-[(2E)-3-[5'-[(1E)-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κN<sup>1</sup>,κN<sup>1</sup>]-2-propenoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-  
**Solvents:** Tetrahydrofuran; 24 h, 115 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

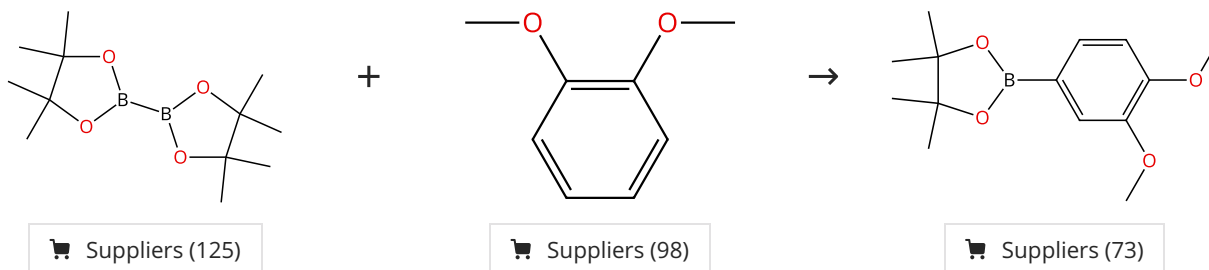
Suppliers (105)

Suppliers (62)

|                                                                                                                                           |                      |                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------|
| 31-142-CAS-17565949                                                                                                                       | Steps: 1 Yield: 100% | <b>Achieving Site Selectivity in Metal-Catalyzed Electron-Rich Carbene Transfer Reactions from N-Tosylhydrazones</b> |
| 1.1 <b>Catalysts:</b> Triphenylarsine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene] di-μ-methoxydiiridium<br><b>Solvents:</b> Octane; 16 h, 120 °C |                      | By: Su, Naijing; et al                                                                                               |
| Experimental Protocols                                                                                                                    |                      | Organic Letters (2017), 19(15), 3990-3993.                                                                           |

Scheme 18 (2 Reactions)

Steps: 1 Yield: 100%

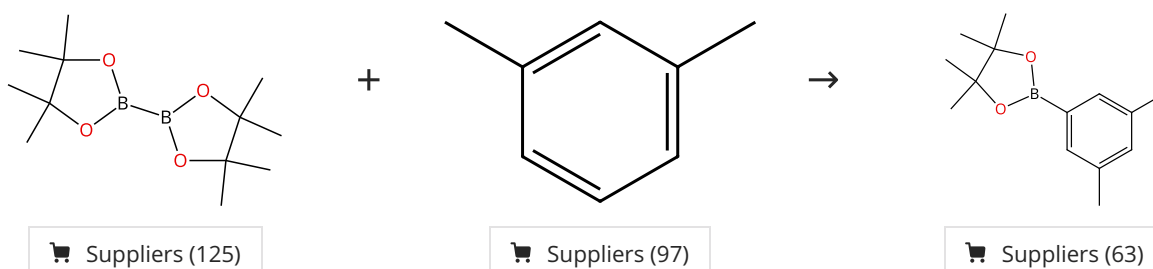


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| 31-142-CAS-4364379                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Steps: 1 Yield: 100% | <b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b> |
| 1.1 <b>Catalysts:</b> Zirconium, tetrakis[μ-[(2 <i>E</i> )-3-[4'-[(1 <i>E</i> )-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κ <i>O</i> :κ <i>O'</i> ]]bis[μ <sub>3</sub> -[(2 <i>E</i> )-3-[5'-[(1 <i>E</i> )-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κ <i>N</i> <sup>1</sup> , κ <i>N</i> <sup>1'</sup> ]-2-propenoato(2-)-κ <i>O</i> :κ <i>O'</i> ]]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetra-μ <sub>3</sub> -oxohexa-<br><b>Solvents:</b> Tetrahydrofuran, Heptane; 36 h, 115 °C |                      | By: Manna, Kuntal; et al                                                                                                                                           |
| Experimental Protocols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      | Journal of the American Chemical Society (2015), 137(7), 2665-2673.                                                                                                |

|                                                                                                                                                                                                                                                                                                                                                          |                      |                                                                                                                                                                    |
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| 31-142-CAS-6502603                                                                                                                                                                                                                                                                                                                                       | Steps: 1 Yield: 100% | <b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b> |
| 1.1 <b>Catalysts:</b> Zirconium, bis[μ <sub>3</sub> -[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κ <i>N</i> <sup>1</sup> , κ <i>N</i> <sup>10</sup> ]benzoato(2-)-κ <i>O</i> :κ <i>O'</i> ]]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetra-μ <sub>3</sub> -oxohexa-<br><b>Solvents:</b> Heptane; 2 d, 115 °C |                      | By: Manna, Kuntal; et al                                                                                                                                           |
| Experimental Protocols                                                                                                                                                                                                                                                                                                                                   |                      | Journal of the American Chemical Society (2015), 137(7), 2665-2673.                                                                                                |

Scheme 19 (2 Reactions)

Steps: 1 Yield: 100%



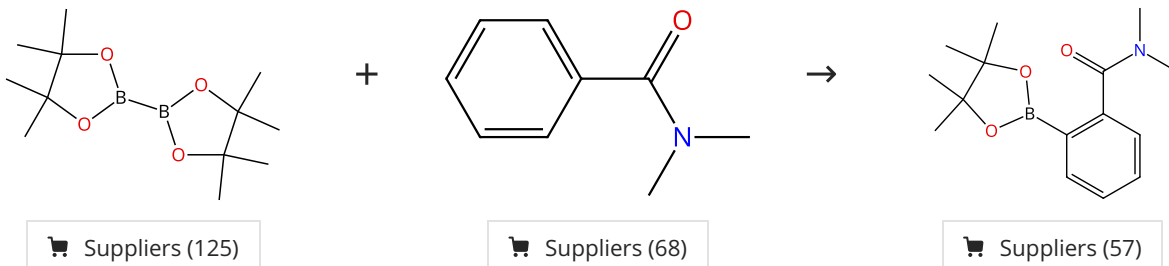
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| 31-142-CAS-6835853                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Steps: 1 Yield: 100% | <b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b> |
| 1.1 <b>Catalysts:</b> Zirconium, tetrakis[μ-[(2 <i>E</i> )-3-[4'-[(1 <i>E</i> )-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κ <i>O</i> :κ <i>O'</i> ]]bis[μ <sub>3</sub> -[(2 <i>E</i> )-3-[5'-[(1 <i>E</i> )-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κ <i>N</i> <sup>1</sup> , κ <i>N</i> <sup>1'</sup> ]-2-propenoato(2-)-κ <i>O</i> :κ <i>O'</i> ]]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ <sub>3</sub> -hydroxytetra-μ <sub>3</sub> -oxohexa-<br><b>Solvents:</b> Tetrahydrofuran; 16 h, 115 °C |                      | By: Manna, Kuntal; et al                                                                                                                                           |
| Experimental Protocols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                      | Journal of the American Chemical Society (2015), 137(7), 2665-2673.                                                                                                |



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| <p><b>31-142-CAS-8968167</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κN<sup>1</sup>,κN<sup>10</sup>]benzoato(2-)-κO:κO']bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetrakis[μ-[2"-nitro[1,1':4',1'':4'',1'''-quaterphenyl]-4,4'''-dicarboxylato(2-)-κO<sup>4'''</sup>:κO<sup>4'''</sup>]]tetra-μ<sub>3</sub>-oxohexa-; 32 h, 115 °C</p> <p>Experimental Protocols</p> | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |
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Scheme 20 (1 Reaction)

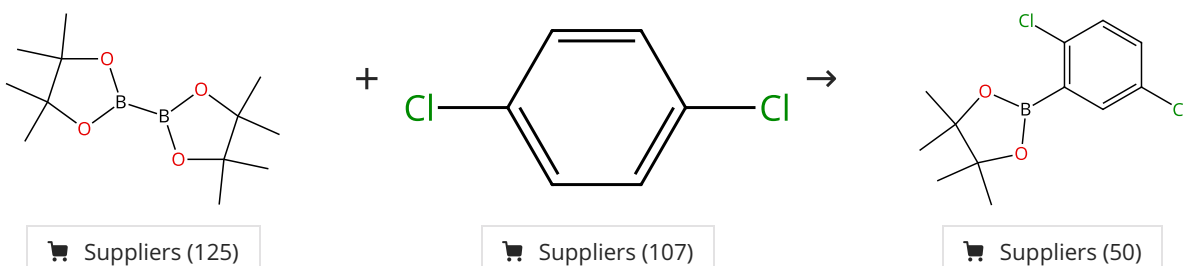
Steps: 1 Yield: 100%



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| <p><b>31-614-CAS-31874942</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 2-[[Bis(1-methylethyl)silyl]methyl]pyridine<br/> <b>Solvents:</b> Tetrahydrofuran; 24 h, 80 °C</p> <p>Experimental Protocols</p> | <p><b>Amide directed iridium C(sp<sup>3</sup>)-H borylation catalysis with high N-methyl selectivity</b></p> <p>By: Dannatt, Jonathan E.; et al</p> <p>Tetrahedron (2022), 109, 132578.</p> |
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Scheme 21 (1 Reaction)

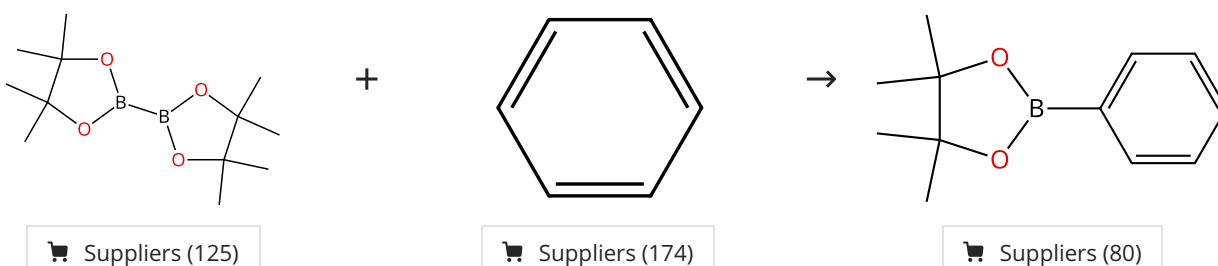
Steps: 1 Yield: 100%



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| <p><b>31-142-CAS-2250241</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κN<sup>1</sup>,κN<sup>10</sup>]benzoato(2-)-κO:κO']bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetrakis[μ-[2"-nitro[1,1':4',1'':4'',1'''-quaterphenyl]-4,4'''-dicarboxylato(2-)-κO<sup>4'''</sup>:κO<sup>4'''</sup>]]tetra-μ<sub>3</sub>-oxohexa-<br/> <b>Solvents:</b> Heptane; 24 h, 115 °C</p> <p>Experimental Protocols</p> | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |
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Scheme 22 (2 Reactions)

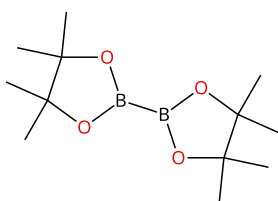
Steps: 1 Yield: 100%



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                      |
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| <p><b>31-142-CAS-9247455</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κN<sup>1</sup>,κN<sup>10</sup>]benzoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetrakis[μ-[2"-nitro[1,1:4',1":4",1"-quaterphenyl]-4,4"-dicarboxylato(2-)-κO<sup>4'''</sup>:κO<sup>4'''</sup>]]tetra-μ<sub>3</sub>-oxohexa-; 18 h, 115 °C</p> <p>Experimental Protocols</p>                                                              | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |
| <p><b>31-142-CAS-7093314</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, tetrakis[μ-[(2<i>E</i>)-3-[4'-[(1<i>E</i>)-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κO:κO']]bis[μ<sub>3</sub>-[(2<i>E</i>)-3-[5'-[(1<i>E</i>)-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κN<sup>1</sup>,κN<sup>1'</sup>]-2-propenoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-<br/><b>Solvents:</b> Tetrahydrofuran; 16 h, 115 °C</p> <p>Experimental Protocols</p> | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |

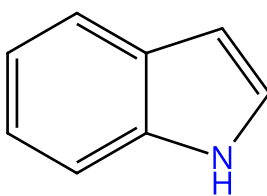
Scheme 23 (3 Reactions)

Steps: 1 Yield: 100%



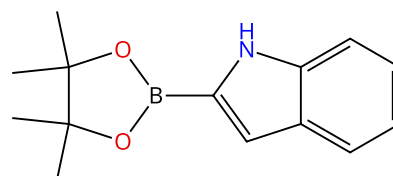
Suppliers (125)

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Suppliers (113)

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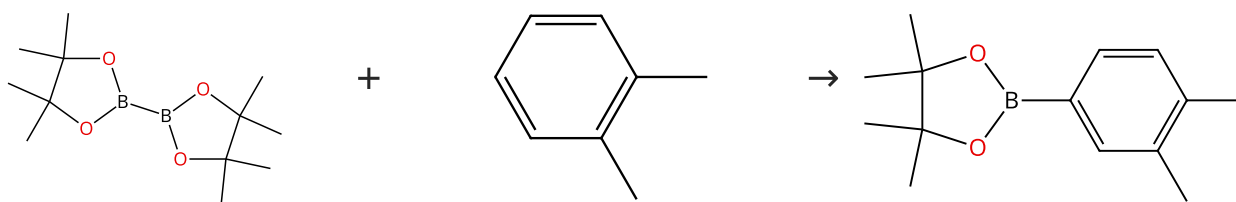


Suppliers (56)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
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| <p><b>31-142-CAS-8635280</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, hexakis[μ<sub>3</sub>-[(2<i>E</i>)-3-[5'-[(1<i>E</i>)-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κN<sup>1</sup>,κN<sup>1'</sup>]-2-propenoato(2-)-κO:κO']]hexakis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-<br/><b>Solvents:</b> Heptane; 18 h, 115 °C</p> <p>Experimental Protocols</p>                                                                                                                    | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |
| <p><b>31-142-CAS-2519363</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, tetrakis[μ-[(2<i>E</i>)-3-[4'-[(1<i>E</i>)-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κO:κO']]bis[μ<sub>3</sub>-[(2<i>E</i>)-3-[5'-[(1<i>E</i>)-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κN<sup>1</sup>,κN<sup>1'</sup>]-2-propenoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-<br/><b>Solvents:</b> Tetrahydrofuran, Heptane; 5 h, 115 °C</p> <p>Experimental Protocols</p> | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |
| <p><b>31-142-CAS-4660708</b> Steps: 1 Yield: 100%</p> <p>1.1 <b>Catalysts:</b> Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthrolin-3-yl-κN<sup>1</sup>,κN<sup>10</sup>]benzoato(2-)-κO:κO']]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetrakis[μ-[2"-nitro[1,1:4',1":4'',1"-quaterphenyl]-4,4''-dicarboxylato(2-)-κO<sup>4'''</sup>:κO<sup>4'''</sup>]]tetra-μ<sub>3</sub>-oxohexa-<br/><b>Solvents:</b> Heptane; 11 h, 115 °C</p> <p>Experimental Protocols</p>                                       | <p><b>Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation</b></p> <p>By: Manna, Kuntal; et al</p> <p>Journal of the American Chemical Society (2015), 137(7), 2665-2673.</p> |

Scheme 24 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (125)

Suppliers (108)

Suppliers (79)

31-142-CAS-11088979

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Zirconium, tetrakis[μ-[(2*E*)-3-[4'-[(1*E*)-2-carboxyethenyl][1,1'-biphenyl]-4-yl]-2-propenoato(2-)-κ*O*:κ*O'*]]bis[μ<sub>3</sub>[(2*E*)-3-[5'-[(1*E*)-2-carboxyethenyl][2,2'-bipyridin]-5-yl-κ*N*<sup>1</sup>, κ*N*<sup>1</sup>]-2-propenoato(2-)-κ*O*:κ*O'*]]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-  
**Solvents:** Tetrahydrofuran; 16 h, 115 °C

Experimental Protocols

**Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation**

By: Manna, Kuntal; et al

Journal of the American Chemical Society (2015), 137(7), 2665-2673.

31-142-CAS-4988059

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Zirconium, bis[μ<sub>3</sub>-[4-[8-(4-carboxyphenyl)-1,10-phenanthroline-3-yl-κ*N*<sup>1</sup>, κ*N*<sup>10</sup>]]benzoato(2-)-κ*O*:κ*O'*]]bis[[[(1,2,5,6-η)-1,5-cyclooctadiene]methoxyiridium]tetra-μ<sub>3</sub>-hydroxytetra-μ<sub>3</sub>-oxohexa-  
**Solvents:** Tetrahydrofuran; 35 h, 115 °C

Experimental Protocols

**Bipyridine- and Phenanthroline-Based Metal-Organic Frameworks for Highly Efficient and Tandem Catalytic Organic Transformations via Directed C-H Activation**

By: Manna, Kuntal; et al

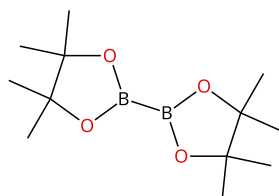
Journal of the American Chemical Society (2015), 137(7), 2665-2673.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component  
structure image  
available in CAS  
SciFinder

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Multi-component  
structure image  
available in CAS  
SciFinder

Suppliers (125)

31-614-CAS-32188639

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 1-Butanaminium, *N,N,N*-tributyl-, *N*-[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)  
**Solvents:** 1,4-Dioxane; 24 h, 50 °C

Experimental Protocols

**para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts**

By: Lu, Shuo; et al

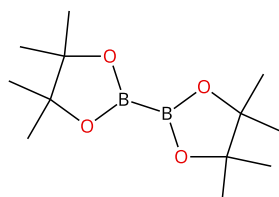
Angewandte Chemie, International Edition (2022), 61(21), e202201285.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component  
structure image  
available in CAS  
SciFinder

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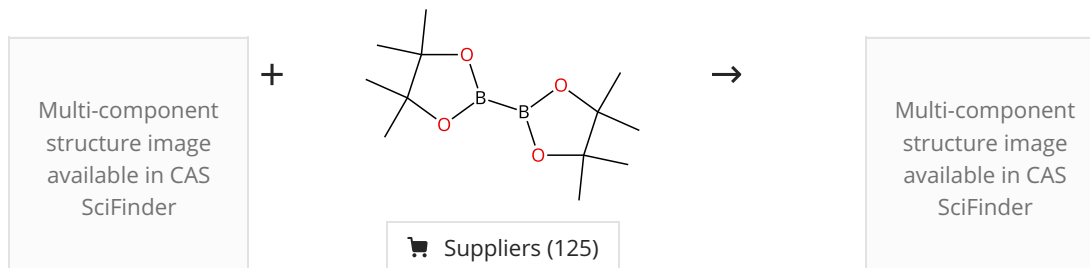
Multi-component  
structure image  
available in CAS  
SciFinder

Suppliers (125)

|                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                           |
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| <b>31-614-CAS-32188638</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenant<br>hrolin-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary<br/>         Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21),<br>e202201285. |
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Scheme 27 (1 Reaction)

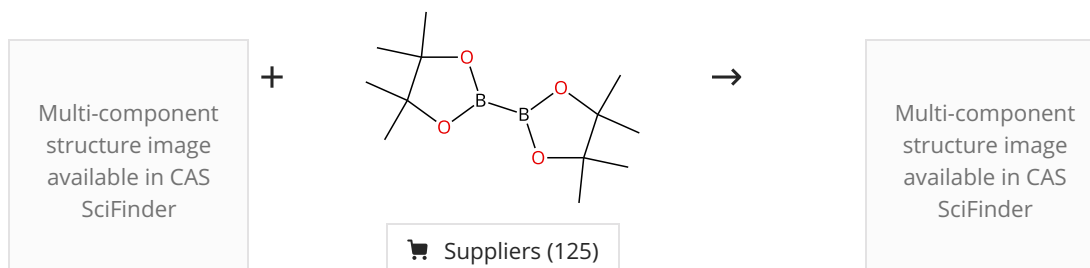
Steps: 1 Yield: 99%



|                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                           |
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| <b>31-614-CAS-32188623</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenant<br>hrolin-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary<br/>         Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21),<br>e202201285. |
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Scheme 28 (1 Reaction)

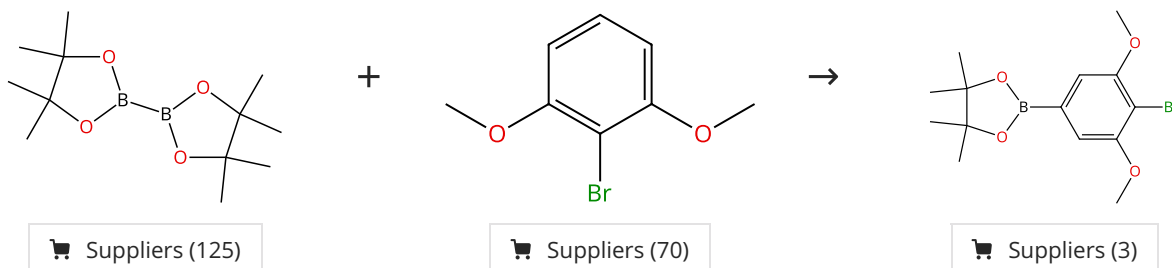
Steps: 1 Yield: 99%



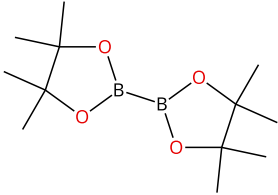
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| <b>31-614-CAS-32188649</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenant<br>hrolin-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary<br/>         Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21),<br>e202201285. |
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Scheme 29 (1 Reaction)

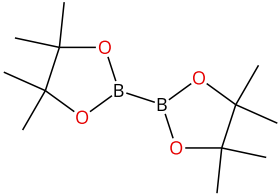
Steps: 1 Yield: 99%



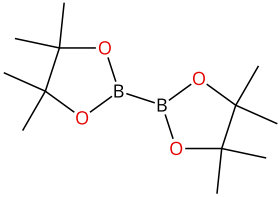
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| <p><b>31-142-CAS-23284513</b> Steps: <b>1</b> Yield: <b>99%</b></p> <p>1.1 <b>Reagents:</b> Oxygen<br/> <b>Catalysts:</b> Isopropanol, Chloro[(1,2,5,6-η)-1,5-cyclooctadiene] (1,10-phenanthroline-κ<sup>N</sup><sup>1</sup>,κ<sup>N</sup><sup>10</sup>)iridium<br/> <b>Solvents:</b> Heptane; 1 h, 75 °C</p> <p>1.2 18 h</p> <p>Experimental Protocols</p> | <p><b>Understanding the Activation of Air-Stable Ir(COD)(Phen)Cl Precatalyst for C-H Borylation of Aromatics and Heteroaromatics</b></p> <p>By: Slack, Eric D.; et al</p> <p>Organic Letters (2021), 23(5), 1561-1565.</p> |
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| <p><b>Scheme 30 (1 Reaction)</b> Steps: <b>1</b> Yield: <b>99%</b></p> <div style="display: flex; align-items: center; justify-content: space-around;"> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> <div>+</div> <div style="text-align: center;">  </div> <div>→</div> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> </div> <div style="text-align: center; margin-top: 10px;">  Suppliers (125)         </div> |  |
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| <p><b>31-614-CAS-32188637</b> Steps: <b>1</b> Yield: <b>99%</b></p> <p>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 1-Butanaminium, <i>N,N,N</i>-tributyl-, <i>N</i>-[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br/> <b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C</p> <p>Experimental Protocols</p> | <p><b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b></p> <p>By: Lu, Shuo; et al</p> <p>Angewandte Chemie, International Edition (2022), 61(21), e202201285.</p> |
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| <p><b>Scheme 31 (1 Reaction)</b> Steps: <b>1</b> Yield: <b>99%</b></p> <div style="display: flex; align-items: center; justify-content: space-around;"> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> <div>+</div> <div style="text-align: center;">  </div> <div>→</div> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> </div> <div style="text-align: center; margin-top: 10px;">  Suppliers (125)         </div> |  |
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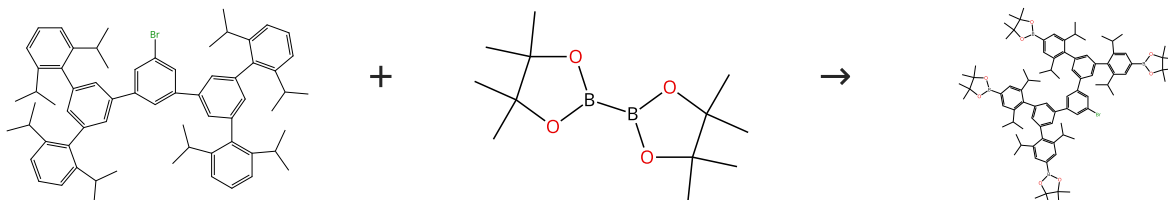
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| <p><b>31-614-CAS-32188648</b> Steps: <b>1</b> Yield: <b>99%</b></p> <p>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 1-Butanaminium, <i>N,N,N</i>-tributyl-, <i>N</i>-[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br/> <b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C</p> <p>Experimental Protocols</p> | <p><b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b></p> <p>By: Lu, Shuo; et al</p> <p>Angewandte Chemie, International Edition (2022), 61(21), e202201285.</p> |
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| <p><b>Scheme 32 (1 Reaction)</b> Steps: <b>1</b> Yield: <b>99%</b></p> <div style="display: flex; align-items: center; justify-content: space-around;"> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> <div>+</div> <div style="text-align: center;">  </div> <div>→</div> <div style="border: 1px solid #ccc; padding: 10px; text-align: center;">             Multi-component structure image available in CAS SciFinder           </div> </div> <div style="text-align: center; margin-top: 10px;">  Suppliers (125)         </div> |  |
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| <b>31-614-CAS-32188645</b> Steps: 1 Yield: 99%<br>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenant<br>hrolin-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21), e202201285. |
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (125)

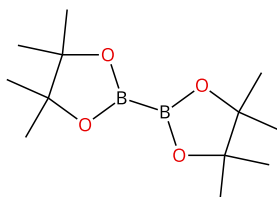
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| <b>31-614-CAS-24637481</b> Steps: 1 Yield: 99%<br>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine<br><b>Solvents:</b> Tetrahydrofuran; 19 h, 80 °C<br>Experimental Protocols | <b>Late-Stage Functionalization of the Periphery of Oligoph<br/>         enylene Dendrimers with Various Arene Units via Fourfold C-<br/>         H Borylation</b><br>By: Masuda, Ryosuke; et al<br>Journal of Organic Chemistry (2021), 86(21), 14433-14443. |
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Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%

 Multi-component  
 structure image  
 available in CAS  
 SciFinder

+



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 Multi-component  
 structure image  
 available in CAS  
 SciFinder

Suppliers (125)

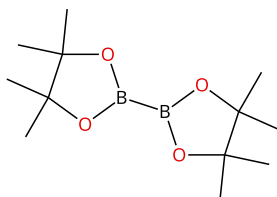
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| <b>31-614-CAS-32188622</b> Steps: 1 Yield: 99%<br>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi<br>iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenant<br>hrolin-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21), e202201285. |
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Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%

 Multi-component  
 structure image  
 available in CAS  
 SciFinder

+



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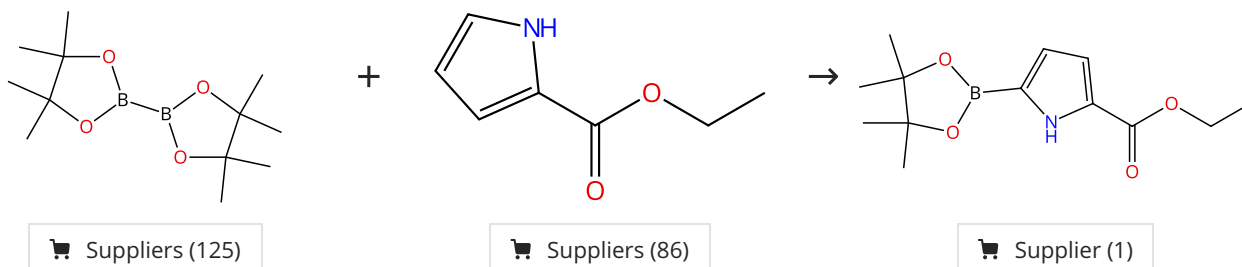
 Multi-component  
 structure image  
 available in CAS  
 SciFinder

Suppliers (125)

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| <b>31-614-CAS-32188624</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi iridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21), e202201285. |
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Scheme 36 (1 Reaction)

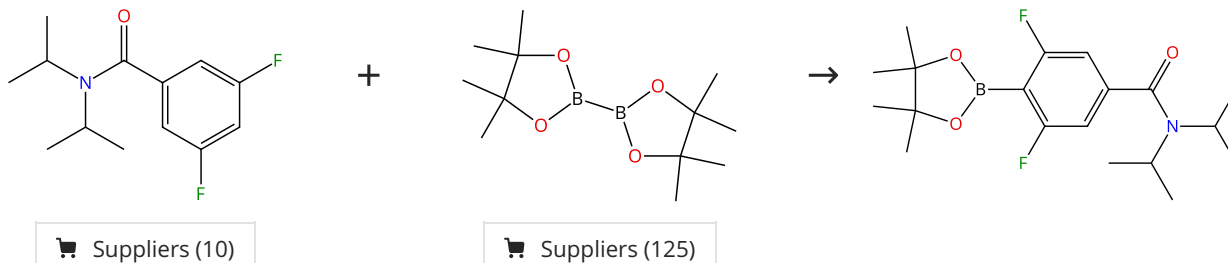
Steps: 1 Yield: 99%



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| <b>31-142-CAS-17236192</b> Steps: 1 Yield: 99%<br><b>1.1 Reagents:</b> Potassium <i>tert</i> -butoxide<br><b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi iridium, 8-[2,2'-Bipyridin]-5-yl-2(1 <i>H</i> )-quinolinone<br><b>Solvents:</b> Tetrahydrofuran; 2 min, rt<br>1.2 12 h, 80 °C<br>Experimental Protocols | <b>Noncovalent interactions in Ir-catalyzed C-H activation: L-shaped ligand for para-selective borylation of aromatic esters</b><br>By: Hoque, Emdadul Md; et al<br>Journal of the American Chemical Society (2017), 139(23), 7745-7748. |
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Scheme 37 (1 Reaction)

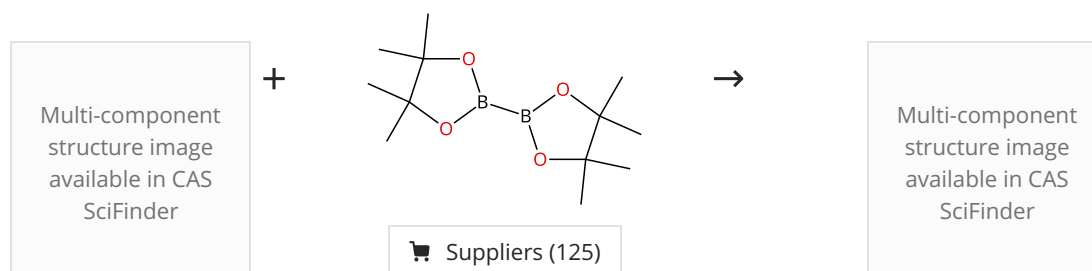
Steps: 1 Yield: 99%



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| <b>31-614-CAS-42482304</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> 1704503-82-1, 1 <i>H</i> -Pyrrolo[2,3- <i>b</i> ]pyridine, 1-[2,2'-bipyridin]-5-yl-<br><b>Solvents:</b> <i>tert</i> -Butyl methyl ether; 5 min, rt<br>1.2 5 min, rt<br>1.3 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydi iridium; 5 min, rt; 12 h, 40 °C | <b>Ligand scaffold for Iridium-catalyzed proximal and remote borylation of arenes</b><br>Assignee: Centre of Biomedical Research<br>World Intellectual Property Organization, WO2024166133 A1 2024-08-15<br><b>PatentPak</b> available |
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Scheme 38 (1 Reaction)

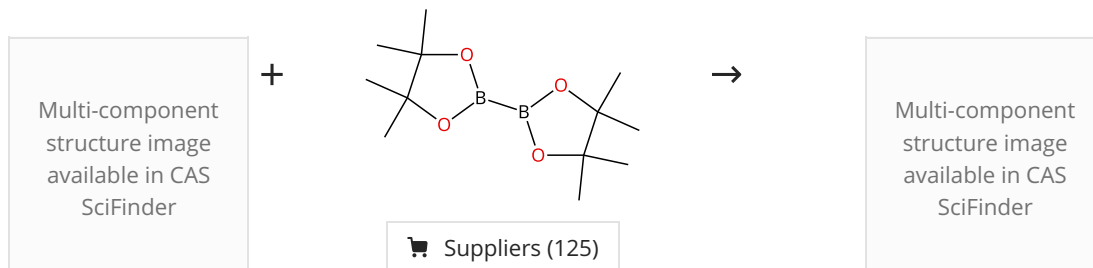
Steps: 1 Yield: 99%



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| <b>31-614-CAS-32188619</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 36 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21), e202201285. |
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Scheme 39 (1 Reaction)

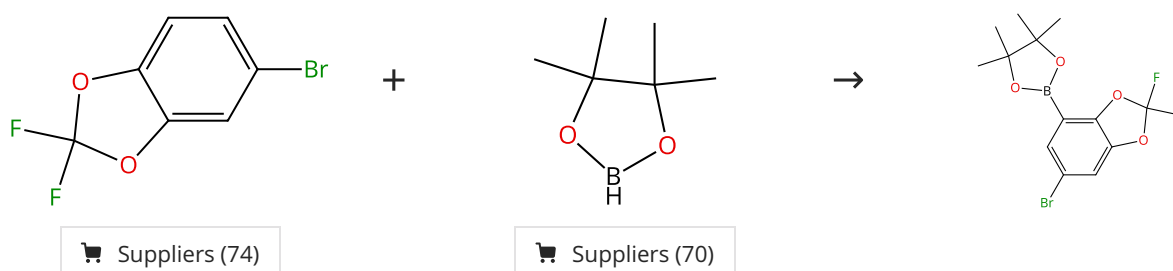
Steps: 1 Yield: 99%



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| <b>31-614-CAS-32188620</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 1-Butanaminium, <i>N,N,N</i> -tributyl-, <i>N</i> -[2'-(1,10-phenanthroline-5-yl)[1,1'-biphenyl]-4-yl]sulfamate (1:1)<br><b>Solvents:</b> 1,4-Dioxane; 24 h, 50 °C<br>Experimental Protocols | <b>para-Selective C-H Borylation of Aromatic Quaternary Ammonium and Phosphonium Salts</b><br>By: Lu, Shuo; et al<br>Angewandte Chemie, International Edition (2022), 61(21), e202201285. |
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Scheme 40 (1 Reaction)

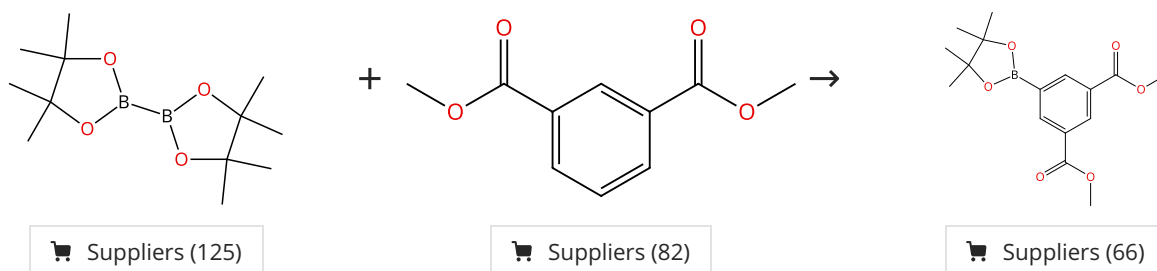
Steps: 1 Yield: 99%



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| <b>31-142-CAS-6425587</b> Steps: 1 Yield: 99%<br><b>1.1 Catalysts:</b> 2,2'-Bipyridine, Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium; 16 h, 80 °C; 80 °C → rt<br>Experimental Protocols | <b>Synthesis of Fluoroalkoxy Substituted Arylboronic Esters by Iridium-Catalyzed Aromatic C-H Borylation</b><br>By: Batool, Farhat; et al<br>Organic Letters (2015), 17(17), 4256-4259. |
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Scheme 41 (1 Reaction)

Steps: 1 Yield: 99%

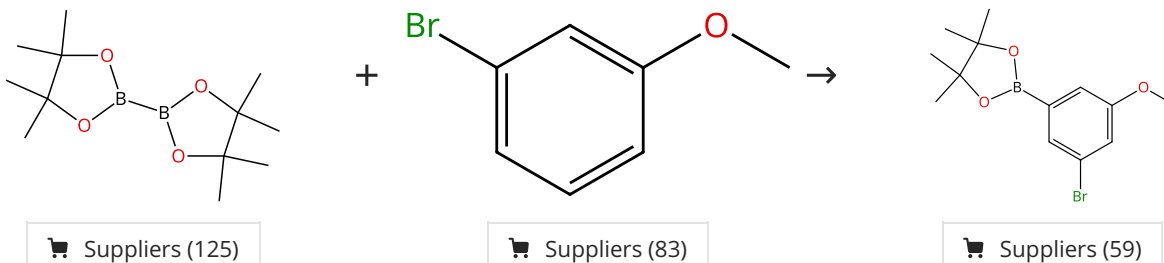




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| <p><b>31-614-CAS-23989366</b> Steps: 1 Yield: 99%</p> <p>1.1 <b>Catalysts:</b> Di-μ-chlorobis[(1,2,5,6-η)-1,5-cyclooctadiene]diiridium, <i>N</i><sup>1</sup>-[4,4'-Bis(1,1-dimethylethyl)[2,2'-bipyridin]-6-yl]-1,2-benzenediamine<br/> <b>Solvents:</b> Cyclopentyl methyl ether; 12 h, 140 °C</p> <p>Experimental Protocols</p> | <p><b>NNB-type tridentate boryl ligands enabling a highly active iridium catalyst for C-H borylation</b></p> <p>By: Ding, Siyi; et al</p> <p>Molecules (2019), 24(7), 1434.</p> |
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Scheme 42 (1 Reaction)

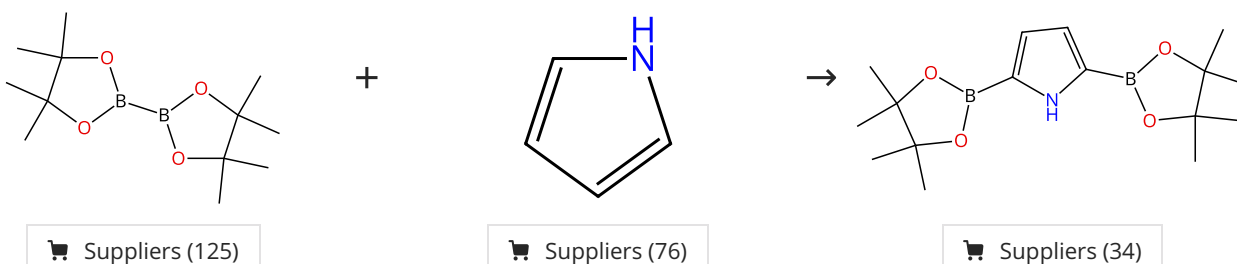
Steps: 1 Yield: 99%



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| <p><b>31-614-CAS-23989367</b> Steps: 1 Yield: 99%</p> <p>1.1 <b>Catalysts:</b> Di-μ-chlorobis[(1,2,5,6-η)-1,5-cyclooctadiene]diiridium, <i>N</i><sup>1</sup>-[4,4'-Bis(1,1-dimethylethyl)[2,2'-bipyridin]-6-yl]-1,2-benzenediamine<br/> <b>Solvents:</b> Cyclopentyl methyl ether; 12 h, 140 °C</p> <p>Experimental Protocols</p> | <p><b>NNB-type tridentate boryl ligands enabling a highly active iridium catalyst for C-H borylation</b></p> <p>By: Ding, Siyi; et al</p> <p>Molecules (2019), 24(7), 1434.</p> |
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Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



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| <p><b>31-142-CAS-17236194</b> Steps: 1 Yield: 99%</p> <p>1.1 <b>Catalysts:</b> Bis[(1,2,5,6-η)-1,5-cyclooctadiene]di-μ-methoxydiiridium, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine<br/> <b>Solvents:</b> Tetrahydrofuran; 2 min, rt</p> <p>1.2 12 h, 80 °C</p> <p>Experimental Protocols</p> | <p><b>Noncovalent interactions in Ir-catalyzed C-H activation: L-shaped ligand for para-selective borylation of aromatic esters</b></p> <p>By: Hoque, Emdadul Md; et al</p> <p>Journal of the American Chemical Society (2017), 139(23), 7745-7748.</p> |
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